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Current job status

- Lecturer, Universitat Politècnica de Catalunya (UPC) (2018-)
- Member and project manager at CITCEA-UPC (2018-)

Previous experience

- Senior engineer, postdoctoral researcher, CITCEA-UPC (2016-2018)
- Adjunct lecturer at UPC (2016-2018)
- Doctoral researcher at UPC/CITCEA UPC (2012-2016)
- Project engineer at CITCEA-UPC (2011-2012)

Education

- Doctor in Electrical Engineering, UPC, May 2016. Excellent cum laude. Title: "Power converter control for offshore wind energy generation and transmission", Supervisors: Oriol Gomis-Bellmunt (UPC), Adrià Junyent-Ferré (Imperial College London).
- KIC InnoEnergy PhD School certificate in Energy Innovation, November 2016.
- Industrial Engineer degree (equivalent to B.S. plus M.S.) specialized in automation and control, School of Industrial Engineering of Barcelona (ETSEIB), March 2011.

Projects

- Participation in competitive and industrial national and international projects.
- Selected competitive projects:
  - "Innovative tools for offshore wind and DC grids" (InnoDC). Innovative Training Network (H2020-ITN). Co-supervision of three doctoral researchers.
  - "Breaking technical, economic and regulatory barriers for the development of DC Supergrids" (SUPERRED). *Retos de colaboración*. Collaboration with Universidad Pontificia de Comillas.
- Selected industrial projects:
  - "Study of control and interaction of the HVDC/FACTS technology in transport grids". Co-supervision of two doctoral researchers in collaboration with Red Eléctrica de España (REE). Funded by REE.
  - "Development of a DC/DC converter for DC Grids". Development of a DC/DC converter to perform studies within a multi-terminal experimental network. Collaboration with: E.C. Lille, Uni. of Aberdeen and RTE. Funded by RTE.

Publications

- 23 journal papers in relevant journals of the sector (IEEE, Elsevier, IET, MDPI, CIGRE)
- Publication of 3 book chapters, 33 conference papers, 4 informative articles.

Patents

1. EP13382510 Harmonics mitigation in multiphase generator-conversion systems. O. Gomis, M. de-Prada, F. Díaz, E. Prieto, ALSTOM RENOVABLES S.L., 13/12/2013
2. EP14382388 Current flow controller. F. Hassan, J. Sau, E. Prieto, O. Gomis, Alstom Grid UK, 10/10/2014
3. EP14382445 Current flow control assembly. F. Hassan, J. Sau, E. Prieto, O. Gomis, Alstom Grid UK, 07/11/2014

International mobility

- Four months at GE Grid Solutions (formerly Alstom Grid UK) (2013), Stafford, UK.
- Short doctoral stays at Imperial College London (2 weeks), London, UK.

Researchers supervision

- Currently supervising 8 doctoral researchers: Joan Sau (2016-2018), Ricard Ferrer (2016-2019), Enric Sánchez (2016-2019) and Carlos Collados (2017-2020), Saman Dadjo (2018-2020), Luis Orellana (2018-2020), Daniel Westerman (2018-2020), J. Montero (2019-2021)

Students supervision

- Supervised Master's Thesis: (8), (2013) March Cheah, (2014) Roger Rifà, (2015) Ricard Ferrer, Gerard Clariana, (2016) Alexandru Sipica, (2017) Joaquín Rebled, Doru Bodgan Bolboceanu, Carlos Collados.
- Supervised Bachelor's Thesis: (18), (2013) Javier Morillo, (2014) Lluc Ulldemolins, Pol Victoria, (2015) Gerard Marcet, Joan Pujol, Pablo Jover, Jordi Romani, Andreu Pagès, Eduard Cirera, (2016) José Montero, (2017) Oriol Portabella, Jordi Sarradell, (2018) Gerard Salmoral, Xavier Gutierrez, Agustí Viladrich, Sergio Gutierrez, Blanca Ràfols, Jaume Girona.

International and national collaborations

- Dr. A. Junyent (Imperial College London, UK), Prof. J. Liang (Cardiff University, UK), Prof. M. Barnes, Oliver Cwikowski, Bin Chang (University of Manchester (UK)), Prof. X Guillaud (École Centrale de Lille, France), Dr. S. Akkari, Dr. J. Dai (Supélec, France), Dr. J. Sakamuri, Dr. N. Cutululis, Dr. A. Hansen, Dr. M. Altin (DTU Wind, Denmark), Dr. D. Van Hertem, Dr. J. Beerten, Dr. A. Bayo (KU Leuven, Belgium), Dr. A. Mokhberdoran, Dr. N. Silva, Prof. A. Carvalho (EFACEC and University of Porto, Portugal), Dr. A. Ferreira (University of Porto, Portugal), Dr. F. Hassan, Dr. S. Fekriasl (GE Grid Solutions, UK), Dr. A. Egea (Strathclyde U, UK), Dr. M. Bongiorno (Chalmers, Sweden), Prof. F. Dörfler, (ETH Zurich, Switzerland), Dr. R. Alves, A. Nami, (ABB, Sweden), N. Degrenne. (Mitsubishi R&D, France), Dr. P. Dworakowski (Supergrid Institute, France), K. Schönleber, Dr. S. Ratés (GE Renewable Energy), Q. Zhao, Dr. J. García, Dr. F. Echavarren (Universidad Pontificia de Comillas), A. Agbemuko, Dr. J.L. Dominguez (IREC), L. Coronado, C. Longás, S. Sanz, A. Cordón (REE, Spain).

Participation in scientific societies

- IEEE Member. Member of the IEEE Working Group 15.05.18 "Studies for Planning of HVDC"
- Cigré Member. Regular member of the WG B4-70 *Guide for Electromagnetic Transient Studies involving VSC converters* (2015-2018). Regular member of TF B4-77 *AC Fault Response Options for VSC HVDC Converters*

Teaching activities

- Courses: Control and signal processing, Wind power: wind turbine control, Electrical machines, Industrial automation and communications, Renewable energy, Electrical energy generation, Grid Integration of renewable energy.

Selected invited presentations

- 2018, Grid integration of renewables at power system level, *Greening the grid CommUnity Event,* Barcelona, Spain.
- 2017, Contribution of HVDC technologies to offshore wind integration, *9th Int. Applied Energy Conf*., Cardiff, UK.
- 2017, Fault response of MMC converters, *IRPWind RI HD-MMC Final Workshop – IRPWIND*, IREC, Barcelona, Spain.
- 2017, Power electronic devices and Converters, HVDC technologies updates, *Cigré HVDC Int. Workshop*, Venice, Italy.
- 2016, Presentation of the "Decentralized Control of a Nine-Phase Permanent Magnet Generator for Offshore Wind Turbines", A*nnual meeting of the Electrical Machinery Committee at the Conf. IEEE PES General Meeting*, Boston, US.
- 2015, How can innovative technologies contribute to the smart energy field? Moving towards the offshore wind power plants. *X Conference, Barcelona Energy Challenges*, Barcelona, Spain.
- 2013, Modular generators for offshore wind turbines, *VIII Int. conference on energy innovation: Wind power to the grid"*, CITCEA-UPC and Endesa, Barcelona, Spain.
- 2011, 2013-2016, Presentations at the annual edition of the *HVDC Colloquium*, Barcelona, Leuven, London, Roskilde, Porto.

Awards

- Second best paper award IEEE Transactions on Energy Conversion of the period 2014-2015 for the paper entitled "Decentralized Control of a Nine-Phase Permanent Magnet Generator for Offshore Wind Turbines". Prize delivered by the "Electrical Machinery Committee" (Authors: Eduardo Prieto-Araujo, Adrià Junyent-Ferré, David Lavèrnia, Oriol Gomis-Bellmunt) during the annual meeting in the conference IEEE PES General Meeting, Boston, 2016.
- Schneider electric best ETSEIB-UPC Projecte Final de Carrera (equivalent to Master's Thesis Electricity/Electronics/Automation) Award, 2011. Title of the project: "Dynamic analysis of multi-terminal VSC-HVDC grids for transporting electrical power from offshore wind farms using direct current."
- Doctoral thesis extraordinary prize in the area of Industrial Engineering 2015/2016 "Power converter control for offshore wind energy generation and transmission"